

ENGR 2405 – Circuits I

Using SciLab for Calculation

Hints and Guidance

SciLab

- Name your file as follows:
 - lastname_firstname_assignment.extension
 - e.g. murthy_purna_lab1.pdf
- Place your file in a folder with a similar name to hold any files you may collect or generate associated with the assignment.
 - This is for your benefit - nothing to submit
- Document your program with comments (//)
 - Your Name
 - Problem number or other reference
 - Explain what you are doing i.e. “// do KCL at node 1

SciLab Hints

- Use built-in constants prefixed with %
 - %pi for 3.14159..., %e = 2.718..., %i for $\sqrt{-1}$
 - You can also define your own constants:
 - %j=%i
 - %tau = 2*%pi
 - %deg2rad = %pi/180
 - The “%” just warns you to not change the value in the body of your program

More SciLab Hints

- Use Help!  Explanations can be terse but still the best way to decipher callable functions
 - Book icon  allows hierarchical contents to be displayed (not as easy)
 - Magnifying icon  allows keyword search (like “%” for built-in constants for example)
 - Place cursor on a function (like complex) and right-click (on Windows) and select “help on complex()” the corresponding help entry will appear.

Program Structure

- Wrap up commonly used code fragments in a function
 - “*function(); code-here; endfunction*” sub-program
 - Example is `cart2polar` and `polar2cart` to convert complex numbers back and forth between Cartesian and polar notation
- To get really fancy, put utility functions in a file and execute that file `exec("<pathname>",-1)` to load the function in that file into your program.
 - Not required but it can simplify multiple copies of your code running around confusing things

SciLab Pitfalls

- Be sure to call trig functions with arguments in *radians*
 - if you are using degrees convert to radians first
- **disp(x)** functions display any type of argument x
 - Adds extra spaces
 - If you are comfortable with C language printf function the SciLab equivalent is mprintf("format",x,y,...)
- Use 2 argument arctangent *atan(y,x)* to get correct 4 quadrant answer otherwise limit your input to $x > 0$
 - As usual result is in radians

Complex Arithmetic

- Initialize complex variable with `complex(x,y)` function
 - Example: `v1 = complex(4,5)`
 - Pick off components of a complex number with `real()` and `imag()` built-in functions as needed
 - You can use complex number with routine algebraic functions (+, -, /, *, ^)
 - Define (complex) matrices
 - Invert as needed [X^{-1} or `inv(X)`]
 - Ordinary algebraic functions also work on matrices

END

Using Simulation and Calculation Tools

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